

MOOSE LAKE CARLTON COUNTY AP, MN, USA

WMO: 727449

Lat: 46.419N

Lon: 92.804W

Elev: 328

StdP: 97.45

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04954

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-25.0	-32.4	0.2	-27.3	-29.8	0.2	-23.6	8.3	-7.0	7.6	-8.0	1.0	320	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	31.0	21.6	28.7	20.7	27.3	19.3	23.1	28.7	21.8	27.0	20.6	25.0	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	16.6	26.7	20.1	15.4	25.2	18.8	14.2	24.0	70.3	28.9	65.1	27.1	60.9	25.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.8	7.0	5.8	-32.4	33.3	3.4	1.9	-34.9	34.7	-36.9	35.8	-38.9	36.9	-41.3	38.3
			-32.7	25.1	3.4	1.2	-35.1	26.0	-37.0	26.7	-38.9	27.4	-41.4	28.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-11.0	-9.2	-2.1	4.9	10.9	16.1	19.8	18.3	14.2	6.9	-0.2	-7.6
	DBStd	11.82	7.60	6.79	6.76	4.89	4.42	3.60	3.18	3.37	4.53	4.52	5.67	6.61
	HDD10.0	2755	652	536	377	163	41	2	0	0	11	120	307	547
	HDD18.3	4949	910	769	633	402	234	83	22	42	138	356	556	805
	CDD10.0	992	0	0	2	12	69	186	303	257	138	23	1	0
	CDD18.3	143	0	0	0	0	4	17	66	41	15	1	0	0
	CDH23.3	1746	0	0	1	10	109	297	733	423	159	13	0	0
	CDH26.7	453	0	0	0	2	27	63	212	113	34	1	0	0
Wind	WSAvg	2.3	2.4	2.3	2.4	2.9	2.6	2.1	1.8	1.7	2.0	2.3	2.4	2.3
Precipitation	PrecAvg	789	21	23	35	66	80	115	115	97	84	79	45	27
	PrecMax	993	58	58	63	195	144	168	191	188	176	175	120	63
	PrecMin	573	0	3	14	15	30	50	37	34	21	14	2	4
	PrecStd	100	15	14	16	38	27	34	36	45	43	42	34	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	8.0	19.9	24.7	30.9	31.3	33.0	32.6	30.9	25.1	17.4	6.1
		MCWB	2.8	4.1	12.2	14.5	18.9	22.3	23.7	22.4	20.3	15.9	10.9	3.0
	2%	DB	2.6	4.1	13.9	20.9	26.3	28.0	31.1	29.8	27.4	21.3	13.0	3.0
		MCWB	0.8	0.9	7.8	11.0	16.0	19.4	22.1	21.2	19.5	14.2	8.7	1.5
	5%	DB	1.0	2.2	9.9	17.4	23.0	27.0	28.9	27.4	25.1	17.5	10.9	2.2
		MCWB	0.1	0.0	5.5	9.1	14.2	18.4	21.3	19.8	18.4	12.1	7.0	1.0
	10%	DB	-0.2	0.2	6.3	14.1	20.9	25.0	27.4	25.9	22.5	14.9	7.6	0.9
		MCWB	-1.0	-1.2	2.4	7.6	12.9	16.9	19.8	18.8	17.1	10.4	5.1	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	4.9	13.3	15.2	20.4	23.7	25.2	24.0	23.0	17.7	12.6	3.8
		MCDB	6.2	8.2	18.4	22.9	26.3	28.8	31.2	30.3	27.3	22.6	15.5	5.1
	2%	WB	1.2	1.9	8.5	12.0	17.8	21.5	23.5	22.4	20.9	15.0	9.6	2.1
		MCDB	2.6	3.9	13.3	18.2	23.3	26.6	29.1	27.8	25.0	19.4	13.3	3.3
	5%	WB	-0.1	0.1	5.5	9.9	15.7	19.7	22.2	21.2	19.6	12.8	7.2	0.8
		MCDB	0.6	1.4	9.3	16.7	21.6	24.7	27.4	25.7	24.0	17.3	9.6	1.5
	10%	WB	-1.5	-1.0	3.3	7.8	13.8	18.1	20.8	19.9	18.0	10.7	5.1	-0.1
		MCDB	-0.6	0.1	5.9	13.6	19.9	22.8	25.2	24.3	21.8	13.4	7.9	0.5
Mean Daily Temperature Range	5% DB	MDBR	10.3	11.1	10.9	11.6	13.4	13.4	13.0	12.6	12.2	10.2	8.7	8.3
		MCDBR	9.6	11.3	14.4	18.3	18.7	16.5	15.3	14.6	15.0	15.6	13.6	8.6
		MCWBR	7.9	8.6	9.0	10.3	9.8	9.0	7.6	7.4	8.2	9.1	8.9	7.2
	5% WB	MCDBR	8.8	10.0	13.4	17.4	16.2	14.6	13.7	13.0	12.6	13.5	12.5	7.1
		MCWBR	7.6	7.9	8.6	10.5	9.8	9.0	7.8	7.3	7.7	8.8	8.7	6.3
Clear-Sky Solar Irradiance	taub	0.247	0.257	0.304	0.356	0.387	0.376	0.380	0.389	0.362	0.312	0.284	0.258	
	taud	2.399	2.382	2.375	2.348	2.311	2.366	2.381	2.375	2.424	2.559	2.521	2.427	
	Ebn at Noon	874	933	926	900	879	889	880	858	852	859	821	815	
	Edh at Noon	68	86	102	116	125	119	116	112	98	73	62	60	
All-Sky Solar Radiation	RadAvg	1.59	2.71	3.82	4.65	5.34	5.82	6.19	5.17	3.91	2.40	1.50	1.12	
	RadStd	0.17	0.23	0.36	0.45	0.41	0.40	0.24	0.40	0.29	0.32	0.15	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page